

NOSS Lab 4 Honeypot Analysis

Seamus Dowling set up a honey pot server to attract hackers (2015). The honey pot ran for several months and sure enough the server attracted a lot of interest from hackers and as a result produced masses of event data. Seamus has kindly supplied us with his log file data to trawl through.

Be warned - there are *millions* of lines of data involved

Our job in this lab session will be to use UNIX commands to analyse the honey pot data and to try and come to some conclusions about the data.

Procedure:

1. Download the Honey Pot data file from my GitHub
2. The file is called hpot.zip. mov it to file in your **Desktop** directory (don't unzip it in the repository it will cause issues when you try pulling the latest versions)
3. The file you downloaded is zipped and compressed, What size is the file (in bytes)?

4. Unzip the file, use the unzip command to do this.
5. The unzip command extracts all the files contained in hpot.zip and places them in a directory called hpot. Change into the hpot directory and take a look at the files (use ls -l)
6. How many files are there? _____
7. Use the man command to find out information on the **du** command. You will need to understand how to use it for the next step or two.
8. What is the combined size of all the files in total in the hpot directory (in bytes)?

9. Calculate the percentage compression - divide the size of the compressed data by the size of the uncompressed data? What answer do you get?

$$\text{Percent Compression} = \frac{\text{Size of Compressed File} * 100\%}{\text{Size of Uncompressed Log Files}}$$

10. How many lines of text are there in all of the log files?

Now we will start analysing the contents of the files in some more detail. We know that the honey pot attracts hacking attempts. Such attempts usually end in failure as shown in the following snippet:

```
2015-01-23 07:09:58+0000 [SSHService ssh-userauth on  
HoneyPotTransport,139295,103.41.124.102] login attempt [root/batman123qwe] failed
```

Very occasionally they succeed as shown below:

```
2015-01-23 07:16:33+0000 [SSHService ssh-userauth on  
HoneyPotTransport,139406,77.236.97.96] login attempt [root/admin] succeeded
```

Your job is to crunch the data and find out the following:

1. How many failed log in attempts were there? _____
2. How many successful log in attempts were there? _____
3. Produce a list of all successful username/password pairs, store it in success.txt
4. Produce a list of all failed username/password pairs, store it in fail.txt
5. The lists you produced in the previous questions have some username/password pairs that are repeated. Produce a sorted list of successful username/password pairs (in alphabetical order, none repeated) and place the list in the file success_sorted.txt
6. Count the number of successful username/password combinations _____
7. Produce a sorted list of failed username/password pairs (in alphabetical order, none repeated) and place the list in the file failed_sorted.txt
8. Count the number of failed username/password combinations _____
9. What percentage of login attempts succeeded? _____